# The Algebraic Landscape of Kochen–Specker Sets in Dimension Three

Michael Kernaghan

February 12, 2026

**Abstract**

We present a systematic computational investigation of the algebraic structures underlying Kochen–Specker (KS) sets in three-dimensional Hilbert space. Through exhaustive surveys of coordinate alphabets drawn from quadratic number fields $\mathbb{Q}(\sqrt{d})$ for $d = 2, \ldots, 30$, cyclotomic fields generated by $n$th roots of unity, and extensions involving the golden ratio, we identify exactly four "algebraic islands" among those tested that support KS sets: the integers (smallest found: 31 vectors), $\mathbb{Q}(\sqrt{2})$ (smallest found: 33), the Eisenstein integers $\mathbb{Z}[\omega]$ (smallest found: 33), and the golden ratio field $\mathbb{Q}(\varphi)$ (smallest found: 52). The last of these is newly discovered. We verify computationally that $n$th roots of unity produce KS-uncolorable sets if and only if $6 \mid n$ for all $n \leq 30$, connecting to the recent result that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality. Using cross-product completion as a classification tool, we show that each island generates a finite closed ray pool with distinct combinatorial structure. We quantify the realizability gap: 49% of random abstract hypergraphs are KS-uncolorable, but none were found realizable in $\mathbb{R}^3$ by our numerical methods, and we show that the Conway–Kochen 31-vector set loses all orthogonalities under 1% coordinate perturbation. We recast the KS theorem as a rounding impossibility—the failure of consistently rounding Born-rule probabilities to deterministic $\{0, 1\}$ valuations—and interpret the gap between the theoretical lower bound of 24 and the constructive upper bound of 31 as measuring the cost of geometric realizability.

## 1 Introduction

The Kochen–Specker (KS) theorem [1] is one of the foundational no-go results of quantum mechanics, demonstrating that quantum observables cannot be assigned pre-existing definite values independently of the measurement context. The theorem is typically proved by exhibiting a finite set of rays (one-dimensional subspaces) in a Hilbert space $\mathcal{H}$ such that no consistent $\{0, 1\}$-valued assignment exists satisfying the constraints imposed by orthogonality.

In dimension $d = 3$, the smallest known KS set is the Conway–Kochen 31-vector set (CK-31), reported by Peres [3] with coordinates drawn from the integer alphabet $\{0, \pm 1, \pm 2\}$. Despite decades of effort, no smaller construction has been found. The best known lower bound is 24 vectors, established by Li, Bright, and Ganesh [13] using SAT solvers with computer-algebra verification (improving the earlier bound of 22 by Uijlen and Westerbaan [9]), leaving a gap of 7 vectors that remains one of the central open problems in KS theory. Trandafir and Cabello [14] recently proved that CK-31 is rigid (unique up to unitary transformation in $\mathbb{C}^3$) and conjectured that 31 is optimal.

Previous computational approaches have primarily searched within fixed coordinate alphabets [2, 7] or generated random abstract hypergraphs and tested realizability [13]. The algebraic structure of the coordinate systems underlying KS sets—specifically, which number fields can support them—has received less systematic attention, though Cortez, Morales, and Reyes [12] proved that $\mathbb{Z}[1/6]$ is the minimal ring extension of $\mathbb{Z}$ exhibiting KS contextuality in dimension 3.

In this paper, we approach the problem from a new direction: we systematically classify the number fields that support KS sets in $\mathbb{R}^3$ and $\mathbb{C}^3$, discovering that the known constructions cluster into exactly four discrete "algebraic islands" among the fields and alphabets tested, separated by algebraic barriers. Our main contributions are:

(i) A systematic survey of quadratic fields $\mathbb{Q}(\sqrt{d})$ for $d = 2, \ldots, 30$, showing that only $d = 2$ supports a KS set (Section 4.3).

(ii) The discovery that the golden ratio field $\mathbb{Q}(\varphi)$ supports a KS set (smallest found: 52 vectors), visible only after cross-product completion of the raw alphabet (Section 8.3).

(iii) Computational evidence that $n$th roots of unity produce KS-uncolorable sets in 3D if and only if $6 \mid n$, verified for all $n \leq 30$ (Section 5.1).

(iv) Quantification of the realizability gap: 49% of random abstract hypergraphs are KS-uncolorable but 0% are realizable in $\mathbb{R}^3$ (Section 7).

(v) A perturbation analysis showing CK-31 is destroyed by 1% coordinate noise (Section 6.2).

(vi) A reinterpretation of the KS theorem as a rounding impossibility, connecting it to the concept of integrality gaps in combinatorial optimization (Section 9).

# 2 Preliminaries

## 2.1 Kochen–Specker sets and colorings

**Definition 1.** *A* ray *in $\mathcal{H} = \mathbb{C}^d$ (or $\mathbb{R}^d$) is a one-dimensional subspace, represented by a nonzero vector up to scalar multiples. Two rays $v, w$ are* orthogonal *if $\langle v|w\rangle = 0$. An* orthogonal triad *in $d = 3$ is a set of three mutually orthogonal rays forming a complete basis.*

**Definition 2.** *A* KS coloring *of a set of rays $S$ is a function $f : S \to \{0, 1\}$ (interpreted as "red" and "green") satisfying:*

*(a)* Orthogonal exclusion*: if $v \perp w$ and $f(v) = 1$, then $f(w) = 0$;*

*(b)* Triad completeness*: for every orthogonal triad $\{v_1, v_2, v_3\} \subset S$, exactly one $v_i$ has $f(v_i) = 1$.*

*A set $S$ is a* Kochen–Specker set *(or* KS-uncolorable*) if no KS coloring exists.*

**Definition 3.** *The* orthogonality graph $G(S)$ *of a ray set* $S$ *has vertices corresponding to rays and edges connecting orthogonal pairs. A* triad *corresponds to a triangle (3-clique) in* $G(S)$*. We use "orthogonality graph" when referring to the pair structure and "orthogonality hypergraph" when emphasizing the triad (3-uniform hyperedge) structure; both encode the same underlying ray configuration.*

## 2.2 Coordinate alphabets and number fields

A *coordinate alphabet* $\mathcal{A} \subset \mathbb{R}$ (or $\mathbb{C}$) is a finite set of values from which ray coordinates are drawn. Given an alphabet $\mathcal{A}$, the induced ray set is

$$S(\mathcal{A}) = \{[v] : v \in \mathcal{A}^3 \setminus \{0\}\}$$

where $[v]$ denotes the equivalence class of $v$ under scalar multiplication (the ray).

The *cancellation identity* of an alphabet is the algebraic equation that enables nontrivial orthogonalities. For the integer alphabet $\{0, \pm 1, \pm 2\}$, the key identity is $1+1 = 2$, which yields the three-term dot-product cancellation $1 \cdot 1 + 1 \cdot 1 + (-1) \cdot 2 = 0$.

## 2.3 Cross-product completion

**Definition 4.** *The* cross-product completion *of a ray set* $S \subset \mathbb{R}^3$ *is the closure* $\bar{S}$ *obtained by iterating: for every orthogonal pair* $v, w \in S$*, add the ray* $[v \times w]$ *to* $S$*, until no new rays are generated. This is well-defined since* $v \perp w$ *implies* $v \times w \neq 0$ *and* $v \times w$ *is orthogonal to both* $v$ *and* $w$*.*

Cross-product completion is the mechanism by which a coordinate alphabet generates its full orthogonality structure. Since cross products are polynomial in the coordinates, completion trivially preserves any number field $K$. The nontrivial property is *finiteness*: for a given alphabet $\mathcal{A}$, the completed ray pool $\bar{S}(\mathcal{A})$ stabilizes after finitely many iterations, and its size and triad structure depend sensitively on the algebraic identities available in $\mathcal{A}$.

# 3 Computational Methods

## 3.1 Ray canonicalization

A ray is an equivalence class of nonzero vectors under scalar multiplication. For *real* alphabets ($\mathcal{A} \subset \mathbb{R}$), two vectors represent the same ray iff they are nonzero real multiples of each other; we choose the primitive representative with first nonzero entry positive. For *complex* alphabets ($\mathcal{A} \subset \mathbb{C}$), two vectors represent the same ray iff they are nonzero complex multiples of each other (i.e., $v \sim \lambda v$ for any $\lambda \in \mathbb{C}^*$); we choose a canonical representative by dividing by the first nonzero coordinate (making it equal to 1), then applying a lexicographic tie-breaker on the remaining coordinates.

In both cases, representatives are *unnormalized*: we do not divide by the vector's norm. This avoids introducing spurious irrationals through normalization and ensures that ray counts (e.g., 49, 109, 205) are well-defined projective counts over the coordinate ring. Orthogonality is tested on the unnormalized representatives: $v \perp w$ iff $\sum_k \bar{v}_k w_k = 0$ (Hermitian, reducing to $\sum_k v_k w_k = 0$ in the real case).

### 3.2 KS coloring via SAT

We encode the KS coloring problem as a Boolean satisfiability (SAT) instance. For a ray set with $n$ rays, orthogonal pairs $E$, and triads $T$:

- Boolean variable $x_i$ indicates ray $i$ is green ($f(i) = 1$).
- For each triad $\{i, j, k\} \in T$: exactly-one constraint (4 clauses).
- For each pair $(i, j) \in E$: at-most-one constraint (1 clause: $\neg x_i \vee \neg x_j$).

We use the Glucose4 solver [10] via the PySAT library. UNSAT = KS-uncolorable.

### 3.3 Randomized greedy minimization

Given a KS-uncolorable set $S$, we find minimal KS subsets by randomized greedy removal: shuffle the rays, attempt to remove each in random order, keeping the removal if the set remains KS-uncolorable. We run 200–1000 independent trials per configuration and report the smallest KS subset found.

### 3.4 Realizability via numerical optimization

Given an abstract orthogonality graph $G$ on $n$ vertices with edge set $E$, we test realizability in $\mathbb{R}^3$ by parametrizing each ray by spherical coordinates $(\theta_i, \phi_i)$ and minimizing

$$\mathcal{L}(\theta, \phi) = \sum_{(i,j)\in E} (v_i \cdot v_j)^2$$

using L-BFGS-B with 50 random restarts. We declare realizability if $\mathcal{L} < 10^{-8}$.

## 4 Real Coordinate Alphabets

### 4.1 Systematic alphabet survey

Table 1 summarizes the results of exhaustive ray generation and KS testing for coordinate alphabets based on various algebraic extensions of $\mathbb{Q}$.

Table 1: KS sets from real coordinate alphabets $\{0, \pm 1, \pm x\}$ in $\mathbb{R}^3$.

| Alphabet | Rays | Pairs | Triads | KS? | Min |
|---|---|---|---|---|---|
| $\{0, \pm 1\}$ | 13 | 24 | 4 | No | — |
| $\{0, \pm 1, \pm\sqrt{2}\}$ | 49 | 120 | 16 | **Yes** | **33** |
| $\{0, \pm 1, \pm 2\}$ | 49 | 138 | 26 | **Yes** | **31** |
| $\{0, \pm 1, \pm\sqrt{3}\}$ | 49 | 114 | 10 | No | — |
| $\{0, \pm 1, \pm\sqrt{5}\}$ | 49 | 114 | 10 | No | — |
| $\{0, \pm 1, \pm\varphi\}$ | 49 | 138 | 10 | No | — |
| $\{0, \pm 1, \pm\frac{1}{2}\}$ | 49 | 138 | 26 | **Yes** | **31** |

The Conway–Kochen 31-vector set is rediscovered by the integer alphabet in 19% of randomized minimization trials (192/1000). No trial produced a set smaller than 31.

## 4.2 The 2:1 ratio as the critical feature

**Observation 5.** *Every real alphabet that achieves a 31-vector KS set contains the ratio* 2:1 *among its nonzero elements—either as* $\pm 2$ *alongside* $\pm 1$*, or equivalently as* $\pm\frac{1}{2}$ *alongside* $\pm 1$*. Adding other algebraic values (e.g.,* $\sqrt{2}$*,* $\varphi$*,* $\sqrt{3}$*) to such an alphabet does not reduce the minimum below 31.*

Table 2 shows extended alphabets. In every case, the presence of the 2:1 ratio yields 31; its absence yields $\geq 33$ or colorable.

Table 2: Extended alphabets: across all tested configurations, the 2:1 ratio correlates perfectly with reaching 31.

| Alphabet | Rays | Triads | Min KS | Has 2:1? |
|---|---|---|---|---|
| $\{0, \pm 1, \pm\sqrt{2}, \pm 2\}$ | 109 | 38 | 31 | Yes |
| $\{0, \pm 1, \pm\varphi, \pm 2\}$ | 145 | 50 | 31 | Yes |
| $\{0, \pm 1, \pm 2, \pm 3\}$ | 145 | 50 | 31 | Yes |
| $\{0, \pm 1, \pm\sqrt{2}, \pm\varphi\}$ | 145 | 28 | 33 | No |
| $\{0, \pm 1, \pm\varphi, \pm\varphi^2\}$ | 109 | 24 | colorable | No |

The cancellation identity $1+1 = 2$ is unique among two-element identities in producing 26 triads from 49 rays—the highest triad density of any two-element alphabet. This density is the mechanism by which the integer alphabet achieves 31. Algebraically, the identity $a + a = b$ with $a, b$ both in the alphabet forces a maximal number of three-term dot-product cancellations $a \cdot a + a \cdot a - b \cdot a = 0$ and its permutations, propagating orthogonalities across coordinate positions. Identities involving irrationals (e.g., $(\sqrt{2})^2 = 2$) produce fewer cancellations because they require squaring to reach an integer, reducing the number of distinct triad-generating permutations.

## 4.3 Quadratic number fields

We systematically tested the alphabet $\{0, \pm 1, \pm\sqrt{d}\}$ for every non-square integer $d = 2, \ldots, 30$.

**Proposition 6.** *Among the quadratic fields* $\mathbb{Q}(\sqrt{d})$ *for* $d \in \{2, 3, 5, 6, 7, \ldots, 30\}$ *(excluding perfect squares), only* $d = 2$ *produces a KS-uncolorable set from the standard two-element alphabet* $\{0, \pm 1, \pm\sqrt{d}\}$*. All others have* $\leq 10$ *triads and are colorable.*

A heuristic explanation: the identity $(\sqrt{d})^2 = d$ produces a three-term dot-product cancellation only when $d$ equals a sum of alphabet elements. For $d = 2$, we have $(\sqrt{2})^2 = 1+1$, giving the Peres cancellation. For $d \geq 3$, no comparable two-term sum from $\{1, \sqrt{d}\}$ equals $d$, so the cancellation structure is too sparse to generate sufficient triads. (A proof that no other cancellation pattern can arise for these alphabets would require a more detailed case analysis, which we leave to future work.)

# 5 Complex Coordinate Alphabets

## 5.1 Roots of unity and the $6 \mid n$ conjecture

We generated ray sets from the alphabet $\{0\} \cup \{\zeta^k : k = 0, \ldots, n-1\}$ where $\zeta = e^{2\pi i/n}$, using the Hermitian inner product $\langle v|w\rangle = \sum_k \bar{v}_k w_k$ for orthogonality. Table 3 shows results for $n = 2, \ldots, 30$.

Table 3: KS sets from $n$th roots of unity. Only multiples of 6 (bold) are uncolorable.

| $n$ | $6 \mid n$? | Rays | Triads | KS? | Min |
|---|---|---|---|---|---|
| 2–5 | No | 13–43 | 1–7 | No | — |
| **6** | **Yes** | **57** | **22** | **Yes** | **33** |
| 7–11 | No | 73–157 | 1–28 | No | — |
| **12** | **Yes** | **183** | **67** | **Yes** | **33** |
| 13–17 | No | 211–343 | 1–76 | No | — |
| **18** | **Yes** | **381** | **136** | **Yes** | **≤76** |
| 19–23 | No | 421–601 | 1–148 | No | — |
| **24** | **Yes** | **651** | **229** | **Yes** | — |
| 25–29 | No | 703–931 | 1–244 | No | — |
| **30** | **Yes** | **993** | **346** | **Yes** | — |

**Conjecture 7.** *The ray set generated by $n$th roots of unity in $\mathbb{C}^3$ is KS-uncolorable if and only if* $6 \mid n$.

This has been verified computationally for all $2 \le n \le 30$ with no exceptions. A proof for general $n$ would likely follow from the representation theory of cyclic groups acting on $\mathbb{C}^3$; we sketch the necessary and sufficient conditions below but leave a complete proof to future work.

The mechanism requires two ingredients working in concert:

- *Three-term phase cancellation* from $3 \mid n$: the identity $1 + \omega + \omega^2 = 0$ (where $\omega = e^{2\pi i/3}$) creates orthogonal triads.
- *Real-axis orthogonalities* from $2 \mid n$: the real embedding ($\zeta^{n/2} = -1$) creates orthogonal pairs that interlock the triads.

Divisibility by 3 alone (e.g., $n = 9, 15, 21, 27$) creates many triads but their constraint network is satisfiable. Divisibility by 2 alone (e.g., $n = 4, 8, 10$) creates pairs but too few triads. Only when both are present ($6 \mid n$) does the interlocking force a KS contradiction.

**Remark 8.** *This result connects to the finding of Cortez, Morales, and Reyes [12] that* $\mathbb{Z}[1/6]$ *is the minimal ring extension of* $\mathbb{Z}$ *exhibiting KS contextuality in dimension 3. The prime factorization* $6 = 2 \times 3$ *appears in both results: the divisibility condition* $6 \mid n$ *in the cyclotomic setting mirrors the necessity of inverting both 2 and 3 in the ring-theoretic setting. This suggests a deep arithmetic structure underlying three-dimensional contextuality.*

### 5.2 Eisenstein integers

The Eisenstein integers $\mathbb{Z}[\omega]$ (where $\omega = e^{2\pi i/3}$) produce KS sets with smallest found subset of 33 vectors across all coefficient bounds tested (max coefficient 1–3, squared norm cutoff 3–7). Increasing the pool size beyond 57 rays does not reduce the smallest found below 33.

## 6 Geometry-First Search

### 6.1 Mixed alphabet completion

We introduce a new search methodology: combine rays from two algebraically independent coordinate alphabets, then close under cross products (Definition 4). The cross products generate rays in *neither* original alphabet, exploring configurations invisible to any single-alphabet search.

Table 4: Mixed alphabet search with cross-product completion.

| Combination | Total rays | Triads | KS? | Min |
|---|---|---|---|---|
| $\{0, \pm1, \pm2\} + \{0, \pm1, \pm\sqrt{2}\}$ | 205 | 158 | Yes | 31 |
| $\{0, \pm1, \pm2\} + \{0, \pm1, \pm\sqrt{3}\}$ | 217 | 164 | Yes | 31 |
| $\{0, \pm1, \pm2\} + \{0, \pm1, \pm\sqrt{5}\}$ | 217 | 164 | Yes | 31 |
| $\{0, \pm1, \pm2\} + \{0, \pm1, \pm\varphi\}$ | 241 | 188 | Yes | 34† |
| $\{0, \pm1, \pm\sqrt{2}\} + \{0, \pm1, \pm\sqrt{3}\}$ | 229 | 166 | Yes | 33 |
| $\{0, \pm1, \pm\sqrt{2}\} + \{0, \pm1, \pm\varphi\}$ | 253 | 190 | Yes | 33 |

Cross-product completion adds 150+ rays beyond the original alphabets, but the minimum KS size does not improve. The 31-vector barrier persists across all algebraically enriched constructions containing the 2:1 ratio. Conversely, combinations lacking the 2:1 ratio (e.g., $\{0, \pm1, \pm\sqrt{2}, \pm\varphi\}$) achieve at best 33 despite their larger pools, because their cancellation identities generate sparser triad networks—confirming that the 2:1 ratio is the rate-limiting algebraic feature, not the pool size.

†The pool $\{0, \pm1, \pm2\} + \{0, \pm1, \pm\varphi\}$ contains all 49 integer-alphabet rays as a subset, so it includes CK-31 as a sub-configuration. The reported minimum of 34 reflects the fact that our randomized greedy minimization (200 trials) did not locate the 31-ray subset within the larger 241-ray pool. On the 49-ray integer-only pool, CK-31 is found in 19% of trials; on a pool five times larger, the hit rate drops below our trial budget.

### 6.2 Perturbation sensitivity

Starting from the exact integer alphabet $\{0, \pm1, \pm2\}$, we add independent Gaussian noise $\mathcal{N}(0, \varepsilon^2)$ to each coordinate before normalization. Two rays $v, w$ are declared orthogonal if $|v \cdot w| < \tau$ with tolerance $\tau = 10^{-6}$. This tolerance is small enough that typical perturbed dot products ($\sim \varepsilon$ for originally orthogonal pairs) are detected as non-orthogonal for $\varepsilon \geq 0.01$.

The key point is that coordinate perturbation does not merely lose orthogonalities through floating-point imprecision—it *genuinely changes* the dot products. For the CK-31 integer vectors, a pair with exact dot product 0 acquires a perturbed dot product of order $\varepsilon$, which for $\varepsilon = 0.01$ is $\sim 0.01$, well above the tolerance $\tau = 10^{-6}$. The orthogonality graph itself changes, not just our ability to detect it.

Table 5: CK-31 under coordinate perturbation ($\tau = 10^{-6}$). Perturbation destroys orthogonalities and the KS property.

| Perturbation $\varepsilon$ | Uncolorable (out of 50) | Smallest found |
|---|---|---|
| 0.00 (exact) | **50/50 (100%)** | **31** |
| 0.01 | 0/50 (0%) | — |
| 0.05 | 0/50 (0%) | — |
| 0.10 | 0/50 (0%) | — |
| 0.20 | 0/50 (0%) | — |
| 0.50 | 0/50 (0%) | — |

**Observation 9.** *The Conway–Kochen set* CK-31 *is not robust under coordinate perturbation: the exact integer relationships are necessary, not merely sufficient. This is ultimately a consequence of the fact that orthogonality in $\mathbb{R}^3$ is an exact algebraic condition (the dot product vanishes), and KS-uncolorability requires a precise interlocking of these exact orthogonalities. Any generic perturbation breaks the interlocking.*

We note that this fragility is, in a sense, expected: KS sets depend on exact orthogonality, and generic perturbations destroy exact orthogonality. The more interesting question is whether *approximate* orthogonality graphs (with some tolerance) can support KS-like uncolorability; this connects to the noise-robust noncontextuality framework of Kunjwal and Spekkens [11], which derives statistical inequalities rather than exact impossibilities. Our perturbation experiment quantifies the sharpness of the transition from the exact to the approximate regime.

This result is complementary to the rigidity theorem of Trandafir and Cabello [14], who proved CK-31 is unique up to unitary transformation. Rigidity concerns the uniqueness of the abstract orthogonality graph's realization; our perturbation result concerns the sensitivity of the graph itself to coordinate noise.

### 6.3 Random seed completion

As a control experiment, we tested whether KS-uncolorable sets can emerge from non-algebraic starting points. Starting with 8–25 random unit vectors (coordinates drawn uniformly from the unit sphere), we tested all pairs for orthogonality with tolerance $\tau = 10^{-6}$, completed any orthogonal pairs via cross products, and iterated until stable. In 0 out of 200 trials did this produce a KS-uncolorable set.

This is largely expected: for continuous random vectors in $\mathbb{R}^3$, exact orthogonal pairs occur with probability zero, and even at tolerance $\tau = 10^{-6}$, the expected number of approximately orthogonal pairs among 25 random rays is $\binom{25}{2} \cdot 2\tau \approx 0.0006$, so virtually no completions occur. The result confirms that KS-uncolorability requires the structured orthogonalities provided by algebraic coordinate systems, not merely a large enough collection of rays.

# 7 The Realizability Gap

We generated random 3-uniform hypergraphs with controlled interlocking (each triad shares at least one ray with another) and tested (a) KS-uncolorability via SAT, then (b) realizability in $\mathbb{R}^3$ via numerical optimization (L-BFGS-B with 50 random restarts; declared realizable if residual $< 10^{-8}$).

**Structural filter.** A necessary condition for realizability in $\mathbb{R}^3$ is that the hypergraph be *linear*: no two distinct triads may share two rays, since in $\mathbb{R}^3$ the third member of a triad is uniquely determined (up to sign) by the other two. Of our 1,800 generated hypergraphs, 38% violated this linearity constraint and are *provably* non-realizable for structural reasons alone. We report realizability results for the full ensemble; restricting to the 1,116 linear instances does not change the outcome (0 found realizable).

**Methodological caveat.** Our realizability test is *heuristic*: failure of the numerical optimizer to find an embedding does not constitute a proof of non-realizability. However, we validated the optimizer on all known realizable KS configurations (CK-31, Peres-33), where it consistently finds embeddings with residuals $< 10^{-10}$, giving confidence that false negatives are rare at these sizes. For the 882 uncolorable hypergraphs, the median best residual (across 50 restarts per instance) was $\mathcal{L} \approx 0.3$, with the 5th percentile at 0.04—orders of magnitude above the $10^{-8}$ threshold, indicating that these are far from realizable rather than borderline failures of optimization.

Table 6: Realizability gap: random abstract hypergraphs. "Found realizable" indicates whether the numerical optimizer found an $\mathbb{R}^3$ embedding.

| Rays | Triads | Tested | Uncolorable | Found realizable |
|---|---|---|---|---|
| 26 | 15 | 200 | 73 (37%) | 0 |
| 26 | 17 | 200 | 125 (63%) | 0 |
| 26 | 19 | 200 | 158 (79%) | 0 |
| 28 | 15–19 | 600 | 301 (50%) | 0 |
| 30 | 15–19 | 600 | 225 (38%) | 0 |
| **Total** | | **1800** | **882 (49%)** | **0** |

**Observation 10.** *The combinatorial problem (find an uncolorable hypergraph) is easy—nearly half of random interlocked hypergraphs are KS-uncolorable. The geometric problem (embed it in $\mathbb{R}^3$) appears to be the bottleneck. Among 882 KS-uncolorable abstract hypergraphs tested, none were found realizable by our numerical optimizer—including those satisfying the linearity constraint necessary for $\mathbb{R}^3$ realizability. This suggests that the space of realizable KS-uncolorable orthogonality graphs is vanishingly rare under our random ensemble. Exact certification of non-realizability (e.g., via Gröbner bases or semidefinite relaxations) for a subset of these cases would strengthen this conclusion and is left to future work.*

# 8 Algebraic Island Classification

## 8.1 Cross-product closure as a classification tool

An important subtlety is that KS behavior depends on the *choice of generating alphabet*, not merely on the ambient number field. For instance, $\mathbb{Q}(\sqrt{5}) = \mathbb{Q}(\varphi)$ as fields, but the alphabet $\{0, \pm 1, \pm\sqrt{5}\}$ produces a colorable set while $\{0, \pm 1, \pm\varphi\}$ (where $\varphi = (1 + \sqrt{5})/2$ is the ring-of-integers generator for $\mathbb{Q}(\sqrt{5})$) produces a KS-uncolorable set after completion. We therefore define:

**Definition 11.** *An* algebraic island *is a pair* $(\mathcal{R}, \mathcal{A})$ *consisting of a subring* $\mathcal{R} \subseteq \mathbb{C}$ *(typically the ring of integers of a number field) and a finite alphabet* $\mathcal{A} \subset \mathcal{R}$ *such that:*

*(i) the ray set* $S(\mathcal{A})$ *generates vectors in* $\mathcal{R}^3$*;*

*(ii) the completion of* $S(\mathcal{A})$ *(see below) terminates in a* finite *ray pool* $\bar{S}$ *with coordinates in* $\mathcal{R}$*;*

*(iii) the completed ray pool* $\bar{S}$ *is KS-uncolorable.*

*Two islands* $(\mathcal{R}_1, \mathcal{A}_1)$ *and* $(\mathcal{R}_2, \mathcal{A}_2)$ *are* equivalent *if their completed ray pools have isomorphic orthogonality hypergraphs.*

The *completion* in condition (ii) depends on whether the island is real or complex:

- **Real islands** ($\mathcal{R} \subset \mathbb{R}$): cross-product completion (Definition 4). For each orthogonal pair $v \perp w$, adjoin $[v \times w]$.
- **Complex islands** ($\mathcal{R} \subset \mathbb{C}$, $\mathcal{R} \not\subset \mathbb{R}$): for each Hermitian-orthogonal pair $v \perp w$ in $\mathbb{C}^3$, adjoin the unique ray $[u]$ spanning $(\text{span}\{v, w\})^\perp$. Concretely, $u$ is the vector of $2 \times 2$ cofactors of the matrix $\begin{pmatrix} \bar{v} \\ \bar{w} \end{pmatrix}$:

$$u_k = (-1)^{k+1} \det \begin{pmatrix} \bar{v}_{k+1} & \bar{v}_{k+2} \\ \bar{w}_{k+1} & \bar{w}_{k+2} \end{pmatrix}, \quad k = 1, 2, 3 \pmod 3.$$

  This is the Hermitian analogue of the cross product. When $\mathcal{R}$ is closed under conjugation, $u \in \mathcal{R}^3$, so completion preserves the coordinate ring.

In $\mathbb{R}^3$, conjugation is trivial and the cofactor formula reduces to the standard cross product, so the two definitions agree.

Note that condition (ii) is automatically satisfied when $\mathcal{R}$ is closed under the coordinate operations of the completion (cross products preserve $\mathcal{R}^3$ for real rings; orthogonal complements preserve $\mathcal{R}^3$ for complex rings closed under conjugation). The *finiteness* of the completed pool and the *combinatorial structure* it generates are the nontrivial properties. The same ring $\mathcal{R}$ may support zero, one, or multiple islands depending on the alphabet.

Table 7: Cross-product closure analysis for *real* islands ($\mathcal{R} \subset \mathbb{R}$). "Compl." = completed ray count; "New mag." = distinct new coordinate magnitudes introduced by completion (e.g., for integers, cross products introduce 3, 4, 5 beyond $\{0, 1, 2\}$); "Min" = smallest KS subset found (200–1000 trials).

| Alphabet $(\mathcal{R}, \mathcal{A})$ | Raw | Compl. | New mag. | 2:1? | Uncol.? | Min |
|---|---|---|---|---|---|---|
| $(\mathbb{Z}, \{0, \pm1, \pm2\})$ | 49 | 109 | +3 | Yes | Yes | **31** |
| $(\mathbb{Z}[\sqrt{2}], \{0, \pm1, \pm\sqrt{2}\})$ | 49 | 145 | +12 | No | Yes | **33** |
| $(\mathbb{Z}[\sqrt{3}], \{0, \pm1, \pm\sqrt{3}\})$ | 49 | 145 | +12 | No | No | — |
| $(\mathbb{Z}[\sqrt{5}], \{0, \pm1, \pm\sqrt{5}\})$ | 49 | 145 | +12 | No | No | — |
| $(\mathbb{Z}[\varphi], \{0, \pm1, \pm\varphi\})$ | 49 | 205 | +21 | No | Yes | **52** |

For the *complex* Eisenstein island, completion uses the Hermitian orthogonal-complement operation (Section 8.1). The Eisenstein alphabet $\{0, \pm1, \pm\omega, \pm\bar{\omega}\}$ with $\omega = e^{2\pi i/3}$ generates 57 rays (22 triads) at max coefficient 1. Complex completion does not expand this pool (all Hermitian-orthogonal complements already lie within it), and the completed set is KS-uncolorable with smallest found subset of 33 vectors.

## 8.2 The four known islands

**Observation 12.** *Among all alphabets and fields tested in our survey, exactly four algebraic islands support KS sets:*

*(1)* ***Integer island*** $(\mathbb{Z}, \{0, \pm1, \pm2\})$*: cancellation* $1+1 = 2$*; smallest found: 31 vectors (*CK-31*).*

*(2)* ***Peres island*** $(\mathbb{Z}[\sqrt{2}], \{0, \pm1, \pm\sqrt{2}\})$*: cancellation* $(\sqrt{2})^2 = 1 + 1$*; smallest found: 33 vectors.*

*(3)* ***Eisenstein island*** $(\mathbb{Z}[\omega], \{0, \pm1, \pm\omega, \pm\bar{\omega}\})$*: cancellation* $1 + \omega + \omega^2 = 0$*; smallest found: 33 vectors.*

*(4)* ***Golden island*** $(\mathbb{Z}[\varphi], \{0, \pm1, \pm\varphi\})$*: cancellation* $\varphi^2 = \varphi + 1$*; smallest found: 52 vectors.*

*No other alphabet from our survey (quadratic fields* $\mathbb{Q}(\sqrt{d})$ *for* $d = 2, \ldots, 30$*; roots of unity with* $6 \nmid n$*; icosahedral symmetry directions) produced a KS-uncolorable set. The hierarchy* $31 < 33 \leq 33 < 52$ *was consistent across 200–1000 randomized minimization trials per configuration; the size distributions are reported in the supplementary data.*

This hierarchy correlates with the algebraic complexity of the cancellation identity:

Table 8: The efficiency hierarchy: simpler cancellations yield smaller KS sets. "Smallest found" reports the best result from randomized greedy minimization (200–1000 trials).

| Rank | Identity | Distinct magnitudes | Triads (raw) | Smallest found |
|---|---|---|---|---|
| 1 | $1 + 1 = 2$ | 2 | 26 | **31** |
| 2 | $(\sqrt{2})^2 = 1 + 1$ | 2 (after squaring) | 16 | **33** |
| $2'$ | $1 + \omega + \omega^2 = 0$ | 1 (complex) | 22 | **33** |
| 3 | $\varphi^2 = \varphi + 1$ | 3 | 10 | **52** |

## 8.3 Discovery of the golden ratio island

The golden ratio field $\mathbb{Q}(\varphi)$ where $\varphi = (1+\sqrt{5})/2$ presents a remarkable case. The raw alphabet $\{0, \pm 1, \pm\varphi\}$ generates 49 rays with only 10 triads—too few for KS-uncolorability. However, cross-product completion generates 156 additional rays (all with coordinates in $\mathbb{Q}(\varphi)$), expanding the pool to 205 rays with 166 triads. This completed set *is* KS-uncolorable, with the smallest KS subset found containing 52 vectors.

Note that $\varphi = (1+\sqrt{5})/2$ is the ring-of-integers generator for $\mathbb{Q}(\sqrt{5})$ (since $5 \equiv 1 \pmod 4$). The alternative alphabet $\{0, \pm 1, \pm\sqrt{5}\}$ (using the field generator rather than the ring-of-integers generator) produces a *different* and colorable set. This demonstrates that the choice of alphabet within a fixed number field is consequential (see Section 8.1).

**Observation 13.** *The golden ratio island is invisible to standard alphabet searches because the raw alphabet is colorable. It becomes KS-uncolorable only through the emergent orthogonalities created by cross-product completion. This suggests there may be other "hidden" islands that require algebraic completion to reveal their KS structure.*

## 8.4 Dead ends

**Icosahedral symmetry.** The icosahedron has 31 symmetry-axis directions—6 vertex, 15 edge, 10 face—a tantalizing coincidence with CK-31. However, these directions produce only 5 triads (colorable). After cross-product completion: 91 rays, 65 triads, still colorable. The reason: the icosahedral group has rotation orders 2, 3, 5 (angles 180°, 120°, 72°) but *no 90° rotations*. The orthogonal group of CK-31 is octahedral, the largest finite subgroup of SO(3) containing 90° rotations.

**Higher quadratic fields.** All $\mathbb{Q}(\sqrt{d})$ for $d = 3, \ldots, 30$ (excluding $d = 4, 9, 16, 25$) produce $\leq 10$ triads and are colorable, even after cross-product completion (for $d \neq 5$; $d = 5$ generates the golden ratio island as $\mathbb{Q}(\sqrt{5}) = \mathbb{Q}(\varphi)$).

# 9 The KS Theorem as a Rounding Impossibility

## 9.1 Fractional vs. integer valuations

The KS coloring problem admits a natural relaxation. For the maximally mixed state $\rho = I/3$, the Born rule assigns each ray $v$ the probability $p(v) = \mathrm{tr}(\rho\,|v\rangle\langle v|) = 1/3$. More generally, for any state $\rho$ and any complete orthonormal basis $\{v_1, v_2, v_3\}$,

$$p(v_1) + p(v_2) + p(v_3) = 1, \qquad p(v_i) \in [0, 1].$$

The KS coloring demands *rounding* these continuous probabilities to $\{0, 1\}$ while preserving the sum: exactly one $v_i$ rounds to 1, the others to 0.

For a single triad, rounding is trivial. The impossibility arises from shared rays: a ray appearing in multiple triads must receive the same rounded value in all of them (non-contextuality). When the sharing structure is sufficiently complex, the rounding constraints form a cycle with no consistent solution.

**Remark 14.** *This is analogous to a* feasibility gap *in combinatorial optimization [15]: the* fractional relaxation *($p(v) \in [0, 1]$ with triad sums equal to 1) is always feasible—the constant assignment $p(v) = 1/3$, corresponding to the maximally mixed state $\rho =$*

*$I/3$, satisfies all constraints. The* integer restriction *($p(v) \in \{0,1\}$) becomes infeasible when the orthogonality graph is complex enough. Unlike a standard integrality gap (which concerns the ratio of optimal values), this is a gap between* feasibility *and* infeasibility*: the fractional problem has solutions while the integer problem has none.*

## 9.2 The hierarchy of cancellation identities

This rounding perspective clarifies a hierarchy:

| Identity | Valuation | KS-blocked? | Physical interpretation |
|---|---|---|---|
| $0+0=0$ | trivial | No | Vacuous (violates sum rule) |
| $\frac{1}{2}+\frac{1}{2}=1$ | fractional $[0,1]$ | **No** | Probabilistic hidden variables |
| $1+1=2$ | sharp $\{0,1\}$ | **Yes** | Deterministic hidden variables |

The KS theorem lives in the gap between $\frac{1}{2}+\frac{1}{2}=1$ (always satisfiable) and $1+1-2=0$ (KS-unsatisfiable at $n \geq 31$). Fractional valuations—corresponding to probabilistic hidden-variable theories—always exist. The impossibility is specific to the demand for deterministic, all-or-nothing certainty. In this sense, the KS theorem is fundamentally about the *impossibility of rounding* $1/3$ to either 0 or 1 consistently across all triads.

## 9.3 The cost of geometric realizability

The rounding framing sharpens the meaning of the gap between the lower bound and CK-31:

- The **lower bound** (24, established by Li, Bright, and Ganesh [13] using SAT solvers with computer-algebra verification, improving the earlier bound of 22 [9]) rules out KS sets with $\leq 23$ rays by exhaustive search over both combinatorial structures and their embeddability in $\mathbb{R}^3$.
- The **upper bound** (31, the Conway–Kochen set) is the smallest explicitly constructed KS set.

The gap between 24 and 31 remains open: the lower bound proof eliminates all candidates with $\leq 23$ rays, while the smallest known construction uses 31. Three features of $\mathbb{R}^3$ geometry constrain which combinatorial structures can be realized:

1. **Forced completions**: if $v \perp w$, then $v \times w$ is the unique (up to sign) third member of any triad containing both. The third ray is determined by geometry, not by choice.
2. **Transitive constraints**: if $v \perp w$ and $v \perp u$, then $w$ and $u$ are confined to the plane perpendicular to $v$, restricting their relative orientations.
3. **Algebraic closure**: cross products of algebraic vectors produce algebraic vectors in the same field. The alphabet forces the orthogonality graph to grow until it stabilizes—for $\{0, \pm 1, \pm 2\}$, the completed pool contains 109 rays, within which the smallest KS-uncolorable subset has 31 rays.

Our perturbation result (Section 6.2) shows this geometric construction is *maximally tight*: the constraint cycle barely closes at 31 rays, and any loosening destroys the KS property.

**Conjecture 15.** *The Conway–Kochen set* CK-31 *achieves the true minimum for KS sets in $\mathbb{R}^3$: no KS set with 30 or fewer rays exists. The gap between the abstract lower bound and 31 reflects the irreducible cost of embedding combinatorial constraints in the geometry of physical space.*

This conjecture is consistent with:

- The rigidity of CK-31 [14] (uniqueness suggests optimality);
- The perturbation knife-edge (Observation 9);
- The absence of any sub-31 KS set across all four algebraic islands (Observation 12);
- The complete failure of 1,800 random hypergraphs to achieve geometric realizability (Section 7).

# 10 Discussion

## 10.1 Connections to ring-theoretic results

The $6 \mid n$ conjecture (Conjecture 7) and the Cortez–Morales–Reyes result that $\mathbb{Z}[1/6]$ is the minimal KS ring [12] both identify the primes 2 and 3 as the arithmetic backbone of three-dimensional contextuality. In the cyclotomic setting, divisibility by 2 provides real-axis orthogonalities and divisibility by 3 provides three-term phase cancellations; both are necessary and jointly sufficient for KS-uncolorability. In the ring-theoretic setting, inverting 2 and 3 is necessary and sufficient for the existence of a KS contextual set of vectors with entries in $\mathbb{Z}[1/N]$. The precise relationship between these two formulations deserves further investigation.

## 10.2 Why four islands?

The four algebraic islands correspond to four fundamentally different cancellation mechanisms:

1. **Integer** ($1 + 1 = 2$): additive doubling of units.
2. **Peres** ($(\sqrt{2})^2 = 2$): squaring annihilates an irrational.
3. **Eisenstein** ($1 + \omega + \omega^2 = 0$): three-term complex phase cancellation.
4. **Golden** ($\varphi^2 = \varphi + 1$): the Fibonacci recurrence.

Each mechanism generates a distinct orthogonality structure in its completed ray set, and the efficiency of the mechanism (measured by the minimum KS set size) correlates inversely with its algebraic complexity. The integer identity involves two distinct magnitudes and produces the densest triad network (26 triads from 49 rays, ratio 0.53). The golden ratio identity involves three distinct magnitudes and produces the sparsest (10 triads from 49 rays, ratio 0.20), requiring completion to 205 rays before becoming KS-uncolorable.

### 10.3 Limitations and open questions

Our search is computational, not exhaustive over all possible number fields. Specifically:

1. **Alphabet choice within fields**: For $d \equiv 1 \pmod 4$, the ring of integers of $\mathbb{Q}(\sqrt{d})$ is $\mathbb{Z}[(1+\sqrt{d})/2]$, not $\mathbb{Z}[\sqrt{d}]$. Our discovery that $\varphi = (1+\sqrt{5})/2$ produces a KS set while $\sqrt{5}$ does not shows that the choice of generator matters. We have tested the ring-of-integers generator $(1+\sqrt{d})/2$ for $d \in \{5, 13, 17, 21, 29\}$; results are reported in the supplementary data.

2. **Cubic and higher extensions**: we have not tested cubic fields $\mathbb{Q}(\sqrt[3]{d})$ or higher-degree algebraic extensions.

3. **Transcendental coordinates**: our methods require algebraic coordinates (for exact orthogonality testing); KS sets with transcendental coordinates are not excluded.

4. **The $6 \mid n$ conjecture**: verified for $n \leq 30$ but not proved for all $n$. A proof would likely follow from the representation theory of cyclic groups acting on $\mathbb{C}^3$.

5. **Minimization is heuristic**: our randomized greedy minimization does not certify optimality. The reported "smallest found" values are upper bounds on the true minimum KS subset size for each island. Exact minimization via integer linear programming (e.g., minimize $|S'|$ subject to $S' \subseteq S$ remaining KS-uncolorable, encoded as a set-cover-like ILP) would certify these bounds. We note that our heuristic is well-calibrated for the integer island: 19% of trials recover CK-31 from the 49-ray pool, and no trial finds anything smaller, consistent with the rigidity result of Trandafir and Cabello [14]. For larger pools (109–205 rays), exact ILP minimization is computationally feasible and would strengthen these results.

6. **Lower bound improvement**: our work does not improve the lower bound of 24; it constrains where a sub-31 construction could come from.

## 11 Conclusion

We have presented a systematic computational investigation of the algebraic landscape underlying Kochen–Specker sets in three dimensions. The principal findings are:

1. Among the number fields and alphabets surveyed, KS sets in $\mathbb{R}^3$ and $\mathbb{C}^3$ cluster into exactly four algebraic islands, with smallest KS subsets found at $31 < 33 \leq 33 < 52$ vectors, corresponding to four distinct cancellation mechanisms.

2. The golden ratio field $\mathbb{Q}(\varphi)$ is a newly discovered island, invisible to raw alphabet searches and revealed only by cross-product completion.

3. The $n$th roots of unity produce KS-uncolorable sets if and only if $6 \mid n$ (verified for $n \leq 30$), connecting cyclotomic number theory to the $\mathbb{Z}[1/6]$ minimality result of Cortez, Morales, and Reyes.

4. The realizability gap appears vast: abstract KS-uncolorability is common (49% of random hypergraphs), but none of 882 uncolorable instances were found realizable by our numerical optimizer.

5. The Conway–Kochen 31-vector set is not robust under coordinate perturbation: 1% noise destroys all exact orthogonalities, consistent with the rigidity result of Trandafir and Cabello.

6. The KS theorem is a rounding impossibility: the failure of consistently rounding Born-rule probabilities to deterministic $\{0, 1\}$ valuations. The 24-to-31 gap measures the cost geometric realizability imposes on abstract combinatorial constraints.

## Reproducibility

All code and data are available at `https://github.com/michaelkernaghan/contextuality` (commit `ddc37b7` corresponds to the tables reported here). The implementation uses Python 3.11 with PySAT 1.8 (Glucose4 solver), NumPy, and SciPy (L-BFGS-B optimizer). Tables 1–7 are generated by `ks_islands.py`; Table 3 by `ks_complex.py`; Table 6 by `ks_geometry.py`. All randomized experiments use `random.seed(42)` for reproducibility.

## Acknowledgments

The author acknowledges the use of Claude (Anthropic) for computational assistance, literature synthesis, and manuscript preparation.

# References

[1] S. Kochen and E. P. Specker, "The problem of hidden variables in quantum mechanics," *J. Math. Mech.* **17**, 59–87 (1967).

[2] A. Peres, "Two simple proofs of the Kochen–Specker theorem," *J. Phys. A: Math. Gen.* **24**, L175–L178 (1991).

[3] A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer Academic Publishers, 1993).

[4] M. Kernaghan, "Bell–Kochen–Specker theorem for 20 vectors," *J. Phys. A: Math. Gen.* **27**, L829–L830 (1994).

[5] M. Kernaghan and A. Peres, "Kochen–Specker theorem for eight-dimensional space," *Phys. Lett. A* **198**, 1–5 (1995).

[6] A. Cabello, J. M. Estebaranz, and G. García-Alcaine, "Bell–Kochen–Specker theorem: A proof with 18 vectors," *Phys. Lett. A* **212**, 183–187 (1996).

[7] M. Pavicic, J.-P. Merlet, B. McKay, and N. D. Megill, "Kochen–Specker vectors," *J. Phys. A: Math. Gen.* **38**, 1577–1592 (2005). arXiv:quant-ph/0409014.

[8] A. Cabello, S. Severini, and A. Winter, "Graph-theoretic approach to quantum correlations," *Phys. Rev. Lett.* **112**, 040401 (2014).

[9] S. Uijlen and B. Westerbaan, "A Kochen–Specker system has at least 22 vectors," *New Generation Computing* **34**, 3–23 (2016).

[10] G. Audemard and L. Simon, "On the Glucose SAT solver," *Int. J. Artif. Intell. Tools* **27**, 1840001 (2018).

[11] R. Kunjwal and R. W. Spekkens, "Beyond the Cabello–Severini–Winter framework: Making sense of contextuality without sharpness of measurements," *Phys. Rev. A* **97**, 052110 (2018).

[12] V. Cortez, R. Morales, and E. J. Reyes, "Minimal ring extensions of the integers exhibiting Kochen–Specker contextuality," arXiv:2211.13216 (2022).

[13] J. Li, C. Bright, and V. Ganesh, "Leveraging SAT solvers for the minimum Kochen–Specker problem," in *Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI-24)*, pp. 1944–1952 (2024). arXiv:2306.13319.

[14] S. Trandafir and A. Cabello, "Two fundamental solutions to the rigid Kochen–Specker set problem and the solution to the minimal Kochen–Specker set problem under one assumption," *Phys. Rev. A* **111**, 052204 (2025). arXiv:2501.11640.

[15] V. V. Vazirani, *Approximation Algorithms* (Springer, 2001).

[16] A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," *J. Math. Mech.* **6**, 885–893 (1957).